

Extension of General Guidelines for PvPlant Site Philippines Digos 28.59MWp

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1.Main S/S

On the main substation, will be installed a Rack 33u with dimensions 24" (w) x 42" (d) x 80" (h) and 4x Signal Patch Panels (SPPs) with dimensions 600mm x 600mm x 250mm. The point installation will be in the equipment room and will be conditioned. Inside the Rack there are 2 SubRacks, a server, a switch type tade and all the necessary equipment. The signal connection of Subracks with the 2 SSPs will be done with 23xLiycy Tp 8 pairs. Each Subrack has its own patch panel.

1.1 Rack

The Rack includes the following materials:

1) Subrack 1:

- A. 1xInaccess RSC10-210 Card (16 Analog Inputs)
- B. 1xInaccess RSC10-545 Card (6 RS-485 Inputs)
- C. 6xInaccess RSC10-300 Card (16 Digital Inputs)

2) Subrack 2:

- A. 7xInaccess RSC10-350 Card (8 Digital Outputs)

3) SERVER

- A. 1xHP ProLiant DL160 Gen9 Server

4) Switch

- A. 1xPlanet FNSW 4800 - switch - 48ports - unmanaged – rack-mountable
- B. 1xSISTM-1040 (16x10/100BASE-TX (RJ-45) to 2x100BASE-FX (SC))

5) UPS

- A. 1xAPC1500BG

6) Power supply

- A. 1x48VDC Axima PSU
- B. 4x24VDC MeanWell PSU

Rack equipment supports the following:

1) Subrack 1:

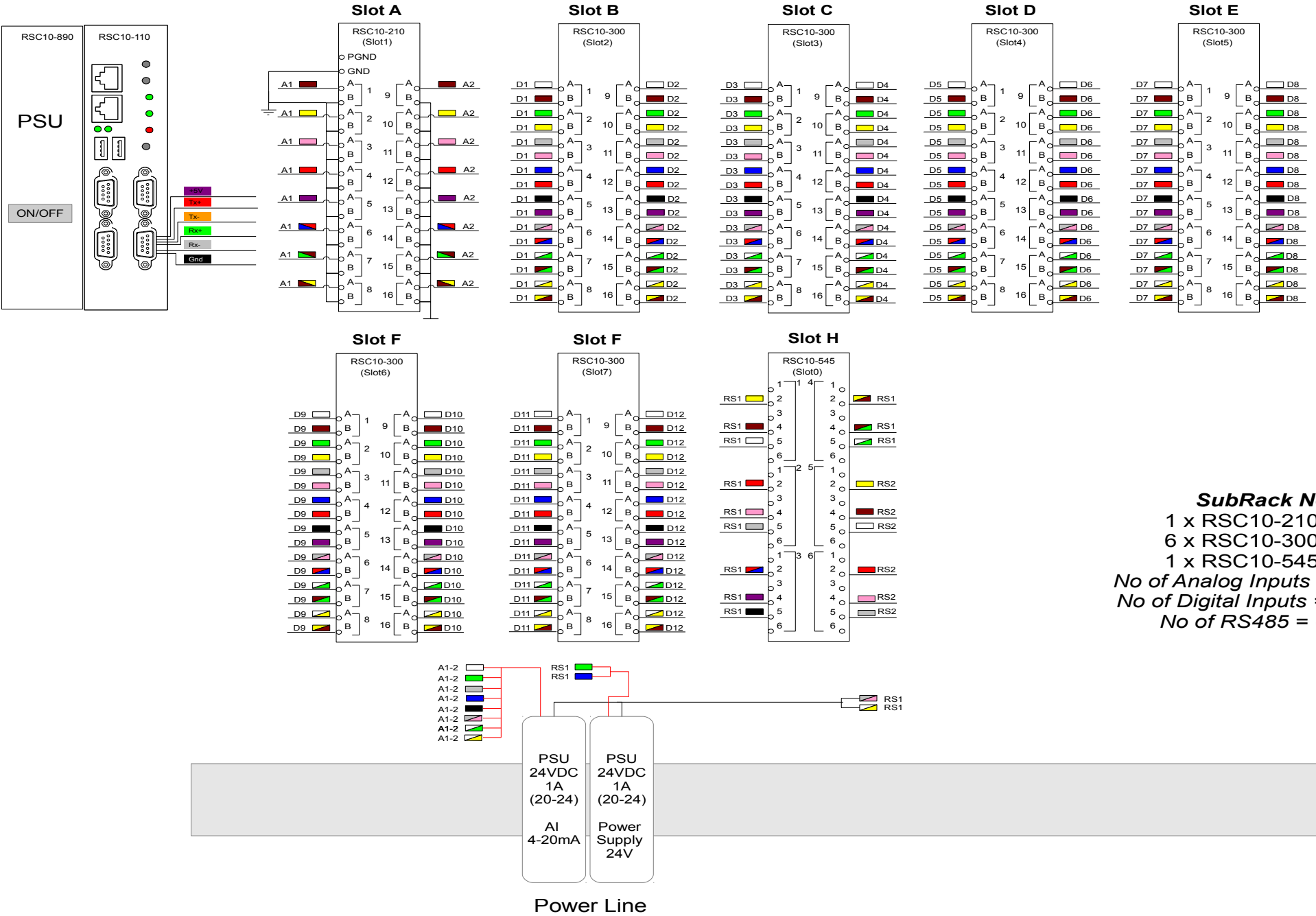
- 1xInaccess RSC10-210 Card (16 Analog Inputs): **0 Analog Inputs**
- 1xInaccess RSC10-545 Card (6 RS-485 Inputs): **2 RS485 Inputs**
- 6xInaccess RSC10-300 Card (16 Digital Inputs): **42 Digital Inputs**

2) Subrack 2:

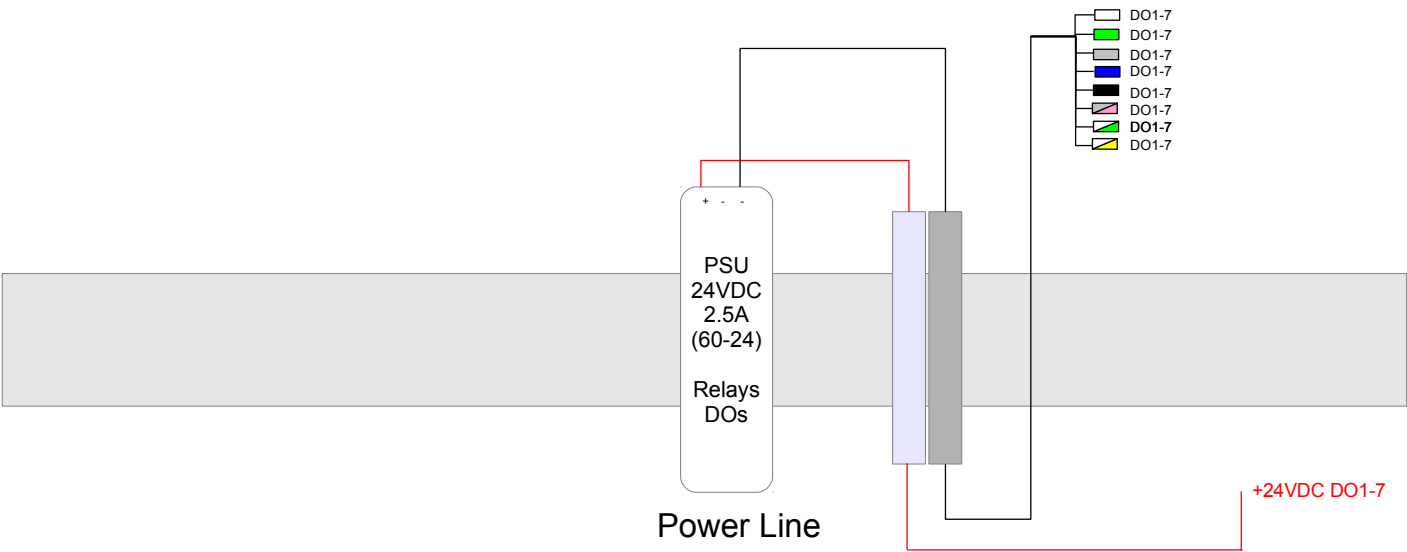
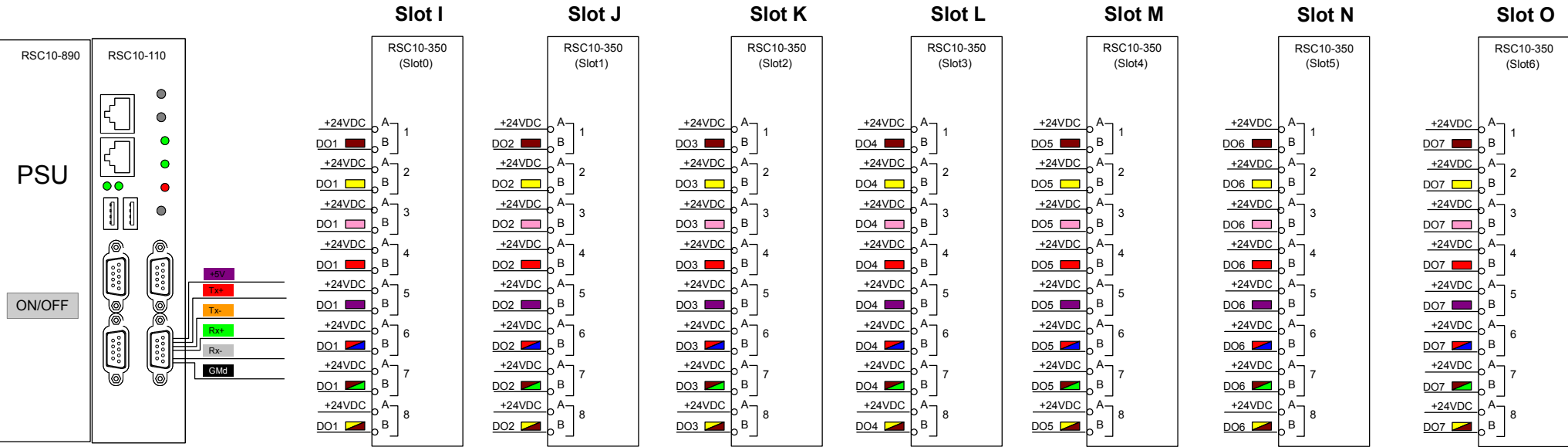
- 7xInaccess RSC10-350 Card (8 Digital Outputs): **28 Digital Outputs**

In the following drawings shown the cabling of 23xLiycy Cables on the side of Rack.

1.1.1 Subrack 1 Detailed Design

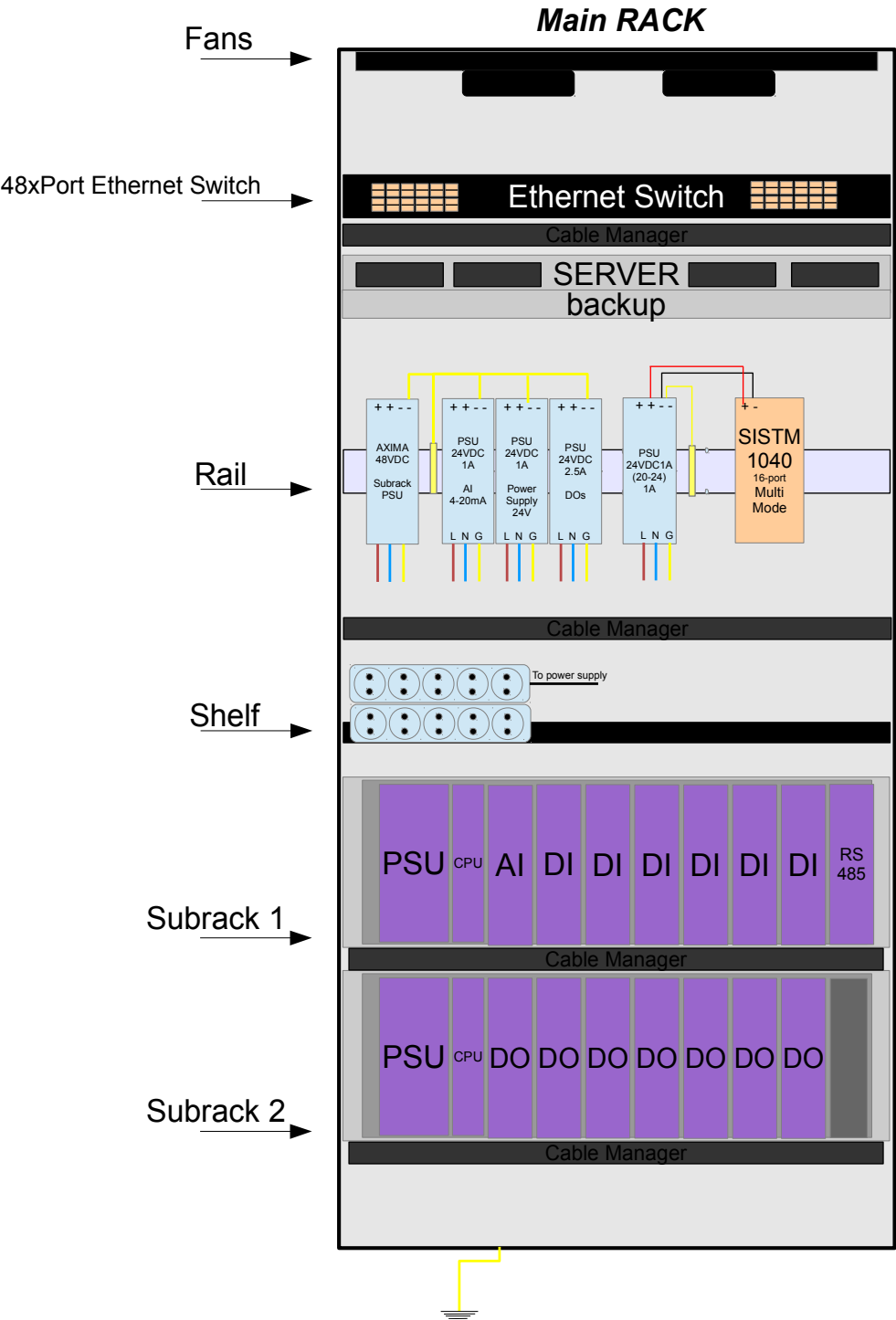


1.1.2 Subrack 2 Detailed Design



SubRack No2
7 x RSC10-350 slots
No of Digital Outputs = 8x7=56

1.1.3 Main Rack Detailed Design



1.2 Signals Patch Panels (SPPs)

To ensure the best possible result in the cabling we provide you 2x SPPs Type Hager to make wiring for all cables that come from the field. Inside the SPPs there are numbered terminals should be connecting the cables of the field to be on the side of terminals that are numbered. There is the appropriate labeling within panels which should be followed.

The following tables shown exactly how to connect the signals from the field as we know.

1.2.1 SPP1 Wiring Table***Signals Patch Panel 1 - Group A***

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
1	Spare	A	1	A1			Slot A : 1
		B	2	A1			
2	Spare	A	3	A1			Slot A : 2
		B	4	A1			
3	Spare	A	5	A1			Slot A : 3
		B	6	A1			
4	Spare	A	7	A1			Slot A : 4
		B	8	A1			
5	Spare	A	9	A1			Slot A : 5
		B	10	A1			
6	Spare	A	11	A1			Slot A : 6
		B	12	A1			
7	Spare	A	13	A1			Slot A : 7
		B	14	A1			
8	Spare	A	15	A1			Slot A : 8
		B	16	A1			
9	Spare	A	17	A2			Slot A : 9
		B	18	A2			
10	Spare	A	19	A2			Slot A : 10
		B	20	A2			
11	Spare	A	21	A2			Slot A : 11
		B	22	A2			
12	Spare	A	23	A2			Slot A : 12
		B	24	A2			
13	Spare	A	25	A2			Slot A : 13
		B	26	A2			
14	Spare	A	27	A2			Slot A : 14
		B	28	A2			
15	Spare	A	29	A2			Slot A : 15
		B	30	A2			
16	Spare	A	31	A2			Slot A : 16
		B	32	A2			
	GND						

Signals Patch Panel 1

Group B

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
1	O/G-1 DC FAIL	A	1	D1			Slot B : 1
		B	2	D1			
2	O/G-1 VCB IN REMOTE	A	3	D1			Slot B : 2
		B	4	D1			
3	O/G-2 DC FAIL	A	5	D1			Slot B : 3
		B	6	D1			
4	O/G-2 VCB IN REMOTE	A	7	D1			Slot B : 4
		B	8	D1			
5	BUSCOUPLER VCB ON	A	9	D1			Slot B : 5
		B	10	D1			
6	BUSCOUPLER VCB SPRING CHARGE	A	11	D1			Slot B : 6
		B	12	D1			
7	BUSCOUPLER VCB SERVICE	A	13	D1			Slot B : 7
		B	14	D1			
8	BUSCOUPLER VCB REMOTE	A	15	D1			Slot B : 8
		B	16	D1			
9	BUSCOUPLER VCB TRIP CKT HEALTHY	A	17	D2			Slot B : 9
		B	18	D2			
10	AUX-TRF-VCB ON	A	19	D2			Slot B : 10
		B	20	D2			
11	AUX-TRF-VCB SERVICE	A	21	D2			Slot B : 11
		B	22	D2			
12	AUX-TRF-VCB REMOTE	A	23	D2			Slot B : 12
		B	24	D2			
13	AUX-TRF-VCB TCH (NEGATE)	A	25	D2			Slot B : 13
		B	26	D2			
14	AUX-TRF-VCB MASTER TRIP OPERATED	A	27	D2			Slot B : 14
		B	28	D2			
15	AUX-TRF-VCB BUCHHOLZ TRIP	A	29	D2			Slot B : 15
		B	30	D2			
16	AUX-TRF-VCB WDG TEMP ALARM	A	31	D2			Slot B : 16
		B	32	D2			
	GND						

Signals Patch Panel 1

Group C

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
17	AUX-TRF-VCB OIL TEMP TRIP	A	1	D3			Slot C : 1
		B	2	D3			
18	AUX-TRF-VCB BUCHHOLZ ALARM	A	3	D3			Slot C : 2
		B	4	D3			
19	AUX-TRF-VCB WDG TEMP ALARM	A	5	D3			Slot C : 3
		B	6	D3			
20	AUX-TRF-VCB OIL TEMP ALARM	A	7	D3			Slot C : 4
		B	8	D3			
21	AUX-TRF-VCB PRV TRIP	A	9	D3			Slot C : 5
		B	10	D3			
22	AUX-TRF-VCB MOG ALARM	A	11	D3			Slot C : 6
		B	12	D3			
23	I/C 1 DC FAIL	A	13	D3			Slot C : 7
		B	14	D3			
24	I/C 1 VCB IN REMOTE	A	15	D3			Slot C : 8
		B	16	D3			
25	I/C 2 DC FAIL	A	17	D4			Slot C : 9
		B	18	D4			
26	I/C 2 VCB IN REMOTE	A	19	D4			Slot C : 10
		B	20	D4			
27	I/C 3 DC FAIL	A	21	D4			Slot C : 11
		B	22	D4			
28	I/C 3 VCB IN REMOTE	A	23	D4			Slot C : 12
		B	24	D4			
29	I/C 4 DC FAIL	A	25	D4			Slot C : 13
		B	26	D4			
30	I/C 4 VCB IN REMOTE	A	27	D4			Slot C : 14
		B	28	D4			
31	I/C 5 DC FAIL	A	29	D4			Slot C : 15
		B	30	D4			
32	I/C 5 VCB IN REMOTE	A	31	D4			Slot C : 16
		B	32	D4			
	GND						

Signals Patch Panel 1

Group D

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
33	I/C 6 DC FAIL	A	1	D5			Slot D : 1
		B	2	D5			
34	I/C 6 VCB IN REMOTE	A	3	D5			Slot D : 2
		B	4	D5			
35	I/C 7 DC FAIL	A	5	D5			Slot D : 3
		B	6	D5			
36	I/C 7 VCB IN REMOTE	A	7	D5			Slot D : 4
		B	8	D5			
37	I/C 8 DC FAIL	A	9	D5			Slot D : 5
		B	10	D5			
38	I/C 8 VCB IN REMOTE	A	11	D5			Slot D : 6
		B	12	D5			
39	I/C 9 DC FAIL	A	13	D5			Slot D : 7
		B	14	D5			
40	I/C 9 VCB IN REMOTE	A	15	D5			Slot D : 8
		B	16	D5			
41	I/C 10 DC FAIL	A	17	D6			Slot D : 9
		B	18	D6			
42	I/C 10 VCB IN REMOTE	A	19	D6			Slot D : 10
		B	20	D6			
43	Spare	A	21	D6			Slot D : 11
		B	22	D6			
44	Spare	A	23	D6			Slot D : 12
		B	24	D6			
45	Spare	A	25	D6			Slot D : 13
		B	26	D6			
46	Spare	A	27	D6			Slot D : 14
		B	28	D6			
47	Spare	A	29	D6			Slot D : 15
		B	30	D6			
48	Spare	A	31	D6			Slot D : 16
		B	32	D6			
	GND						

Signals Patch Panel 1

Group E

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
49	Spare	A	1	D7			Slot E : 1
		B	2	D7			
50	Spare	A	3	D7			Slot E : 2
		B	4	D7			
51	Spare	A	5	D7			Slot E : 3
		B	6	D7			
52	Spare	A	7	D7			Slot E : 4
		B	8	D7			
53	Spare	A	9	D7			Slot E : 5
		B	10	D7			
54	Spare	A	11	D7			Slot E : 6
		B	12	D7			
55	Spare	A	13	D7			Slot E : 7
		B	14	D7			
56	Spare	A	15	D7			Slot E : 8
		B	16	D7			
57	Spare	A	17	D8			Slot E : 9
		B	18	D8			
58	Spare	A	19	D8			Slot E : 10
		B	20	D8			
59	Spare	A	21	D8			Slot E : 11
		B	22	D8			
60	Spare	A	23	D8			Slot E : 12
		B	24	D8			
61	Spare	A	25	D8			Slot E : 13
		B	26	D8			
62	Spare	A	27	D8			Slot E : 14
		B	28	D8			
63	Spare	A	29	D8			Slot E : 15
		B	30	D8			
64	Spare	A	31	D8			Slot E : 16
		B	32	D8			
	GND						

Signals Patch Panel 1

Group F

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
65	Spare	A	1	D9			Slot F : 1
		B	2	D9			
66	Spare	A	3	D9			Slot F : 2
		B	4	D9			
67	Spare	A	5	D9			Slot F : 3
		B	6	D9			
68	Spare	A	7	D9			Slot F : 4
		B	8	D9			
69	Spare	A	9	D9			Slot F : 5
		B	10	D9			
70	Spare	A	11	D9			Slot F : 6
		B	12	D9			
71	Spare	A	13	D9			Slot F : 7
		B	14	D9			
72	Spare	A	15	D9			Slot F : 8
		B	16	D9			
73	Spare	A	17	D10			Slot F : 9
		B	18	D10			
74	Spare	A	19	D10			Slot F : 10
		B	20	D10			
75	Spare	A	21	D10			Slot F : 11
		B	22	D10			
76	Spare	A	23	D10			Slot F : 12
		B	24	D10			
77	Spare	A	25	D10			Slot F : 13
		B	26	D10			
78	Spare	A	27	D10			Slot F : 14
		B	28	D10			
79	Spare	A	29	D10			Slot F : 15
		B	30	D10			
80	Spare	A	31	D10			Slot F : 16
		B	32	D10			
	GND						

Signals Patch Panel 1

Group G

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Starting Point	Slot : Channel
81	Spare	A	1	D11			Slot G : 1
		B	2	D11			
82	Spare	A	3	D11			Slot G : 2
		B	4	D11			
83	Spare	A	5	D11			Slot G : 3
		B	6	D11			
84	Spare	A	7	D11			Slot G : 4
		B	8	D11			
85	Spare	A	9	D11			Slot G : 5
		B	10	D11			
86	Spare	A	11	D11			Slot G : 6
		B	12	D11			
87	Spare	A	13	D11			Slot G : 7
		B	14	D11			
88	Spare	A	15	D11			Slot G : 8
		B	16	D11			
89	Spare	A	17	D12			Slot G : 9
		B	18	D12			
90	Spare	A	19	D12			Slot G : 10
		B	20	D12			
91	Spare	A	21	D12			Slot G : 11
		B	22	D12			
92	Spare	A	23	D12			Slot G : 12
		B	24	D12			
93	Spare	A	25	D12			Slot G : 13
		B	26	D12			
94	Spare	A	27	D12			Slot G : 14
		B	28	D12			
95	Spare	A	29	D12			Slot G : 15
		B	30	D12			
96	Spare	A	31	D12			Slot G : 16
		B	32	D12			
	GND						

Signals Patch Panel 1
Group H

No	Signal description	Contact	Terminal Block	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	WMS comm	RS-485 D +	1	RS1			Slot H: 1
		RS-485 D -	2	RS1			
		GND	3	RS1			
2	UPS-1	RS-485 D +	4	RS1			Slot H : 2
		RS-485 D -	5	RS1			
		GND	6	RS1			
3	Spare	RS-485 D +	7	RS1			Slot H: 3
		RS-485 D -	8	RS1			
		GND	9	RS1			
4	Spare	RS-485 D +	10	RS1			Slot H: 4
		RS-485 D -	11	RS1			
		GND	12	RS1			
5	Spare	RS-485 D +	13	RS2			Slot H: 5
		RS-485 D -	14	RS2			
		GND	15	RS2			
6	Spare	RS-485 D +	16	RS2			Slot H: 6
		RS-485 D -	17	RS2			
		GND	18	RS2			

1.2.2 SPP2 Wiring Table***Signals Patch Panel 2******Group I***

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Starting Point	Slot : Channel
1	O/G-1 VCB OPEN	A	1	11	DO1			Slot I : 1
		B		14	DO1			
2	O/G-1 VCB CLOSE	A	2	11	DO1			Slot I : 2
		B		14	DO1			
3	O/G-2 VCB OPEN	A	3	11	DO1			Slot I : 3
		B		14	DO1			
4	O/G-2 VCB CLOSE	A	4	11	DO1			Slot I : 4
		B		14	DO1			
5	BUSCOUPLER VCB OPEN	A	5	11	DO1			Slot I : 5
		B		14	DO1			
6	BUSCOUPLER VCB CLOSE	A	6	11	DO1			Slot I : 6
		B		14	DO1			
7	AUX-TRF-VCB OPEN	A	7	11	DO1			Slot I : 7
		B		14	DO1			
8	AUX-TRF-VCB CLOSE	A	8	11	DO1			Slot I : 8
		B		14	DO1			
	GND							

Signals Patch Panel 2

Group J

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	I/C 1 VCB OPEN	A	1	11	DO2			Slot J : 1
		B		14	DO2			
2	I/C 1 VCB CLOSE	A	2	11	DO2			Slot J : 2
		B		14	DO2			
3	I/C 2 VCB OPEN	A	3	11	DO2			Slot J : 3
		B		14	DO2			
4	I/C 2 VCB CLOSE	A	4	11	DO2			Slot J : 4
		B		14	DO2			
5	I/C 3 VCB OPEN	A	5	11	DO2			Slot J : 5
		B		14	DO2			
6	I/C 3 VCB CLOSE	A	6	11	DO2			Slot J : 6
		B		14	DO2			
7	I/C 4 VCB OPEN	A	7	11	DO2			Slot J : 7
		B		14	DO2			
8	I/C 4 VCB CLOSE	A	8	11	DO2			Slot J : 8
		B		14	DO2			
	GND							

Signals Patch Panel 2

Group K

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	I/C 5 VCB OPEN	A	1	11	DO3			Slot K : 1
		B		14	DO3			
2	I/C 5 VCB CLOSE	A	2	11	DO3			Slot K : 2
		B		14	DO3			
3	I/C 6 VCB OPEN	A	3	11	DO3			Slot K : 3
		B		14	DO3			
4	I/C 6 VCB CLOSE	A	4	11	DO3			Slot K : 4
		B		14	DO3			
5	I/C 7 VCB OPEN	A	5	11	DO3			Slot K : 5
		B		14	DO3			
6	I/C 7 VCB CLOSE	A	6	11	DO3			Slot K : 6
		B		14	DO3			
7	I/C 8 VCB OPEN	A	7	11	DO3			Slot K : 7
		B		14	DO3			
8	I/C 8 VCB CLOSE	A	8	11	DO3			Slot K : 8
		B		14	DO3			
	GND							

Signals Patch Panel 2

Group L

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	I/C 9 VCB OPEN	A	1	11	DO4			Slot L : 1
		B		14	DO4			
2	I/C 9 VCB CLOSE	A	2	11	DO4			Slot L : 2
		B		14	DO4			
3	I/C 10 VCB OPEN	A	3	11	DO4			Slot L : 3
		B		14	DO4			
4	I/C 10 VCB CLOSE	A	4	11	DO4			Slot L : 4
		B		14	DO4			
5	Spare	A	5	11	DO4			Slot L : 5
		B		14	DO4			
6	Spare	A	6	11	DO4			Slot L : 6
		B		14	DO4			
7	Spare	A	7	11	DO4			Slot L : 7
		B		14	DO4			
8	Spare	A	8	11	DO4			Slot L : 8
		B		14	DO4			
	GND							

Signals Patch Panel 2
Group M

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	Spare	A	1	11	D05			Slot M : 1
		B		14	D05			
2	Spare	A	2	11	D05			Slot M : 2
		B		14	D05			
3	Spare	A	3	11	D05			Slot M : 3
		B		14	D05			
4	Spare	A	4	11	D05			Slot M : 4
		B		14	D05			
5	Spare	A	5	11	D05			Slot M : 5
		B		14	D05			
6	Spare	A	6	11	D05			Slot M : 6
		B		14	D05			
7	Spare	A	7	11	D05			Slot M : 7
		B		14	D05			
8	Spare	A	8	11	D05			Slot M : 8
		B		14	D05			
	GND							

Signals Patch Panel 2
Group N

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Starting Point	Slot : Channel
1	Spare	A	1	11	DO6			Slot N : 1
		B		14	DO6			
2	Spare	A	2	11	DO6			Slot N : 2
		B		14	DO6			
3	Spare	A	3	11	DO6			Slot N : 3
		B		14	DO6			
4	Spare	A	4	11	DO6			Slot N : 4
		B		14	DO6			
5	Spare	A	5	11	DO6			Slot N : 5
		B		14	DO6			
6	Spare	A	6	11	DO6			Slot N : 6
		B		14	DO6			
7	Spare	A	7	11	DO6			Slot N : 7
		B		14	DO6			
8	Spare	A	8	11	DO6			Slot N : 8
		B		14	DO6			
	GND							

Signals Patch Panel 2
Group 0

No	Signal description	Contact	Relay	Relay Input (NO)	Controller cable name	Controller cable color	Staring Point	Slot : Channel
1	Spare	A	1	11	D07			Slot 0 : 1
		B		14	D07			
2	Spare	A	2	11	D07			Slot 0 : 2
		B		14	D07			
3	Spare	A	3	11	D07			Slot 0 : 3
		B		14	D07			
4	Spare	A	4	11	D07			Slot 0 : 4
		B		14	D07			
5	Spare	A	5	11	D07			Slot 0 : 5
		B		14	D07			
6	Spare	A	6	11	D07			Slot 0 : 6
		B		14	D07			
7	Spare	A	7	11	D07			Slot 0 : 7
		B		14	D07			
8	Spare	A	8	11	D07			Slot 0 : 8
		B		14	D07			
	GND							

1.3 Rack and SPPs Interconnection

The interconnection of the SPPs with the RACK should be done from below the panel through the seal which have already installed. The cables from the field will be right to enter from above the SPPs.

2. Peripheral Substations

There are 13 Peripheral Substations. Also there is a Substation for PPC.

2.1 Panels include the following materials

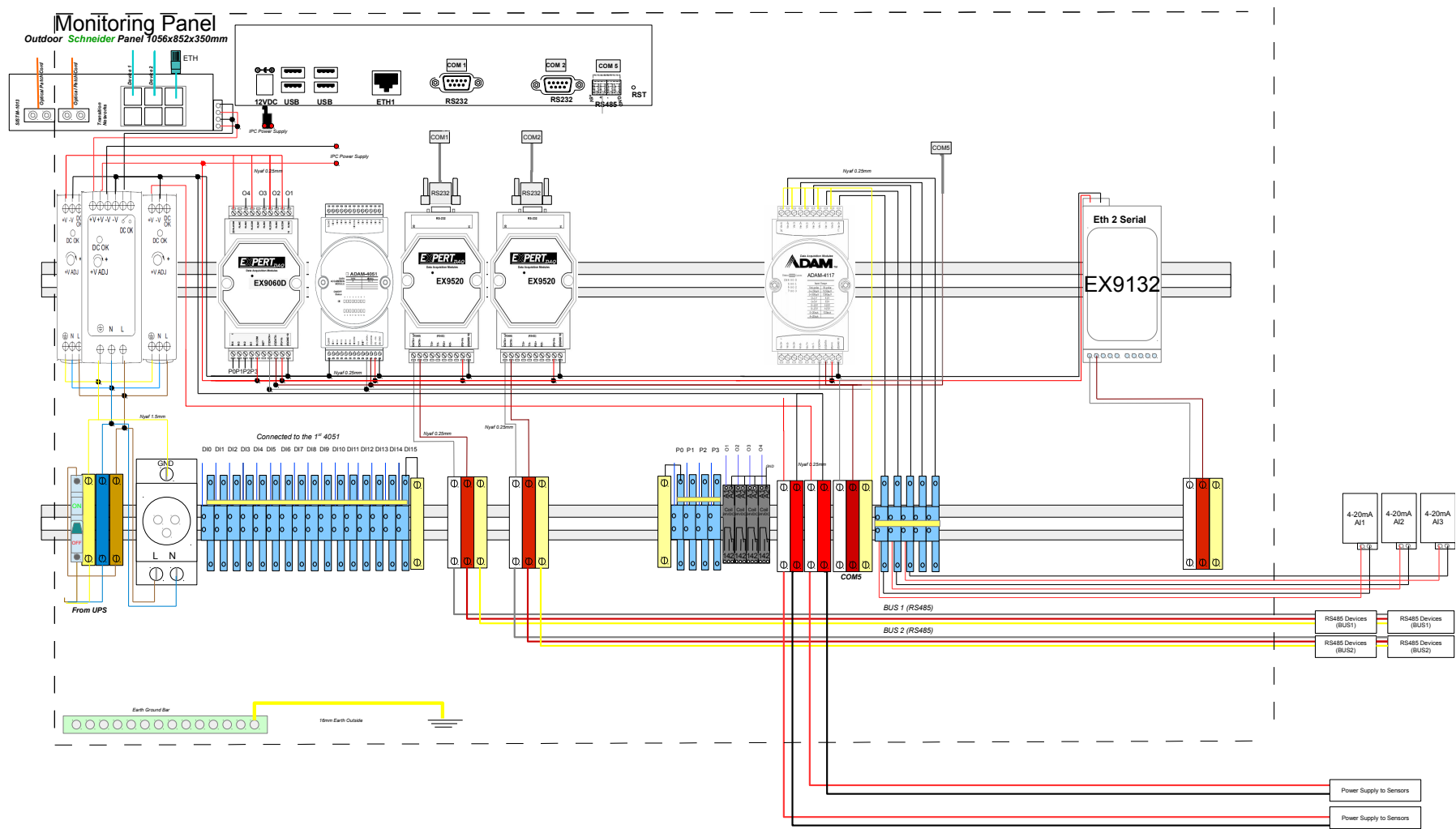
1. 1xInaccess Industrial Controller RSC23 (2xRS232, 1xRS485, 2xUSB, 1xLanEthernet)
2. 2xExpert Topscoc 9520-R Topscoc Converter RS232 to RS485
3. 1x4051 Adam Advantech 16xDIs
4. 1xExpert Topscoc 9060 4DIs, 4DOs
5. 1xAdam Advantech 4117 (8xAnalog Inputs 4-20mA, 0-15V, 0-1V)
6. 1x12VDC3.33A PSU Meanwell (40-12)
7. 2x24VDC1.0A PSU Meanwell (20-24)
8. 1xExpert Topscoc 9132 Eth2serial
9. 1xSISTM-1013 (6x10/100BASE-TX (RJ-45) to 2x100BASE-FX (SC))

2.2 Panels support the following

1. Meters and Protection Devices
2. Several digital contacts from MV cells, AUX T/F, T/F, Security Alarms.
3. Meteo Station – Pyranometers
4. Room and Module Temperature
5. Inverters
6. String Combiner Boxes
7. Analog Sensors

2.3 Connection Diagram of PSSU

The following drawings show the Connection Diagram of PSSU.



inaccess
inAccess S.A
Sorou 12, Marousi Athens
15125 Tel: +30 2106802300
Fax: +30 2106199258

Drawing Title:
V6Rs
Monitoring System Single Diagram

Project Name & Address:

Status:

Approval Stamp:

Optical Fiber
RS232 Cable DB9
RJ45 Cable
Lycy 2x2x0.5mm Cable D+
Lycy 2x2x0.5mm Cable D-
Lycy 2x2x0.5mm Cable GND
Lycy 4x2x0.5mm Cable
V/mA
0.25mm Digital Contact
0.25mm +/-24VDC

LEGEND

2.4 Wiring Table of PSSU for S/S 1-10, 12-13

#	Signal Description	Starting Point	Contact	Cable Color	Comments/Channels
Digital Inputs Adam 4051 (ID:3)					
X	GND			black	
1	RMU CLOSE				DI:0
2					
3	RMU OPEN				DI:1
4					
5	RMU EPS				DI:2
6					
7	RMU ES				DI:3
8					
9	RMU-VCB BUCHHOLZ ALARM				DI:4
10					
11	RMU-VCB WDG TEMP ALARM				DI:5
12					
13	RMU-VCB OIL TEMP ALARM				DI:6
14					
15	RMU-VCB MOG ALARM				DI:7
16					
17	RMU-VCB PRD ALARM				DI:8
18					
19	Spare				DI:9
20					
21	Spare				DI:10
22					
23	Spare				DI:11
24					
25	Spare				DI:12
26					
27	Spare				DI:13
28					
29	Spare				DI:14
30					
31	Spare				DI:15
32					
Controller Serial ports					
B				black	
1	DPX-V5 RELAY		+		COM 1
2			-		IDs:
	GND			black	
3	SMB LOOP-1 (INVERETR-1) &	SCBs	+		COM 2

	SMB LOOP-2 (INVERETR-2)				
4			-		IDs:
	GND			black	
5	Modules/Spare	Sensors	+		COM 5
6			-		IDs: 1,2,3
	GND			black	
7	WMS		+		COM 3
8	(Expert 9132)		-		IDs:
Analog Inputs Expert 4117 (ID:1)					
A	GND			black	
1	24Volt-DC output		+		SCB's Supply Voltage
2			-		
3	24Volt-DC output		+		Meteo, Sensors, Pyranometers Supply Voltage
4			-		
5	MODULE TEMP	PT100 (4-20mA)	+		AI:0
6			-		
7	Spare	4-20mA	+		AI:1
8			-		
9	Spare	4-20mA	+		AI:2
10			-		
11	Spare	4-20mA	+		AI:3
12			-		
13	Spare	4-20mA	+		AI:4
14			-		
Digital Inputs/Outputs Expert 9060D (ID:2) - Pulse					
Z	GND			black	
1	Spare	PI	+		PI:0
2			-		
3	Spare	PI	+		PI:1
4			-		
5	Spare	PI	+		PI:2
6			-		
7	Spare	PI	+		PI:3
8			-		
9	Spare	Output – Relay 24V	A1 (+)		RL:1
10			A2 (-)		
11	Spare	Output – Relay 24V	A1 (+)		RL:2
12			A2 (-)		
13	Spare	Output – Relay 24V	A1 (+)		RL:3
14			A2 (-)		
15	Spare	Output – Relay 24V	A1 (+)		RL:4
16			A2 (-)		
	GND			black	

2.5 Wiring Table of PSSU for S/S 11

#	Signal Description	Starting Point	Contact	Cable Color	Comments/Channels
Digital Inputs Adam 4051 (ID:3)					
X	GND			black	
1	RMU CLOSE				DI:0
2					
3	RMU OPEN				DI:1
4					
5	RMU EPS				DI:2
6					
7	RMU ES				DI:3
8					
9	RMU-VCB BUCHHOLZ ALARM				DI:4
10					
11	RMU-VCB WDG TEMP ALARM				DI:5
12					
13	RMU-VCB OIL TEMP ALARM				DI:6
14					
15	RMU-VCB PRD ALARM				DI:7
16					
17	RMU-VCB MOG ALARM				DI:8
18					
19	RMU CLOSE				DI:9
20					
21	RMU OPEN				DI:10
22					
23	RMU EPS				DI:11
24					
25	RMU ES				DI:12
26					
27	RMU-VCB BUCHHOLZ ALARM				DI:13
28					
29	RMU-VCB WDG TEMP ALARM				DI:14
30					
31	RMU-VCB OIL TEMP ALARM				DI:15
3					

2					
Controller Serial ports					
B				black	
1	DPX-V5 RELAY & DPX-V5 RELAY		+		COM 1
2			-		IDs:
	GND			black	
3	SMB LOOP-1 (INVERETR-1) & SMB LOOP-2 (INVERETR-2)		+		COM 2
4			-		IDs:
	GND			black	
5	Modules/Spare	Sensors	+		COM 5
6			-		IDs: 1,2,3
	GND			black	
7	SMB LOOP-3 (INVERETR-3) & WMS		+		COM 3
8	(Expert 9132)		-		IDs:
Analog Inputs Expert 4117 (ID:1)					
A	GND			black	
1	24Volt-DC output		+		SCB's Supply Voltage
2			-		
3	24Volt-DC output		+		Meteo, Sensors, Pyranometers Supply Voltage
4			-		
5	MODULE TEMP	PT100 (4-20mA)	+		AI:0
6			-		
7	Spare	4-20mA	+		AI:1
8			-		
9	Spare	4-20mA	+		AI:2
10			-		
11	Spare	4-20mA	+		AI:3
12			-		
13	Spare	4-20mA	+		AI:4
14			-		
Digital Inputs/Outputs Expert 9060D (ID:2) - Pulse					
Z	GND			black	
1	RMU-VCB PRD ALARM	PI	+		PI:0
2			-		
3	RMU-VCB MOG ALARM	PI	+		PI:1
4			-		
5	Spare	PI	+		PI:2
6			-		
7	Spare	PI	+		PI:3
8			-		

9	Spare	Output – Relay 24V	A1 (+)		RL:1
10			A2 (-)		
11	Spare	Output – Relay 24V	A1 (+)		RL:2
12			A2 (-)		
13	Spare	Output – Relay 24V	A1 (+)		RL:3
14			A2 (-)		
15	Spare	Output – Relay 24V	A1 (+)		RL:4
16			A2 (-)		
	GND			black	

3.2 Features

Inaccess PPC functionality integrates in a unique manner Control, Monitoring and Grid Integration:

- Active and reactive power closed-loop control with ramp rate and grid limitations: Control of P, Q, cosphi based on static, dynamic setpoints or grid code curves integrating centralized ramp rate control and grid code limitations (Q-V, P-Q).
- Grid integration with any protocol, digital and analogue interface, fully extensible and customizable: Standard IP protocols (DNP3, IEC 60870-5-101, IEC 60870-5-104, Modbus TCP), analogue inputs (0-20mA, 4-20mA, 0-10V) or Digital inputs and outputs
- Support for nearly any market inverter, meter and protection device, on demand support of new devices, reactive power compensation devices and battery systems.
- Interface of any available monitored plant and device measurements to the grid, from generation and compensation assets, meteorological instruments, circuit breakers and other equipment.
- Circuit breakers control, automation of substations disconnection and connection: for user operations or compliance with grid code requirements (transformers premagnetization automation)
- Integration of project-specific, new grid code or plant operator requirements upon request, customized for any plant design or network architecture: Ensuring on-time integration while satisfying any project-specific user or grid operator requirement.
- Interoperable with Inaccess monitoring system for enhanced grid integration monitoring: Offering intelligent user interface, monitoring, visualization, exporting and reporting services, as well as monitoring of grid integration.
- Support for advanced system and network redundancy schemes are integrated and provided upon request

3.3 Control functionality

The PPC system offers flexible closed-loop control functionality easily adaptable to regional requirements to address the grid control needs. Core control functionality supported includes the following.

Active Power Control P	Active power control with dynamic or fixed setpoints, analogue or discrete power levels
Ramp Rate Control	Control the active power up and down ramp rates, dynamic or fixed
Reactive Power Control Q, Q(P), Q night	Reactive power control, dynamic, fixed Q setpoints or Q(P) power curve
Power Factor Control $\cos\phi$, $\cos\phi(P)$	Power factor control, high accuracy, dynamic, fixed or $\cos\phi(P)$
Voltage Stabilization Q(V), $\cos\phi(V)$ Q(V), $\cos\phi(V)$	Voltage-dependent reactive power control using grid code curves
Frequency Support P(f) P(f)	Frequency-dependent active power control using grid code curves
Circuit Breakers control	Distributed control, gradual substations/transformers connection or disconnection, transformers premagnetization automation
Start/Stop Plant Generation	Without disconnecting from the grid

The active/reactive power can be controlled in discrete steps or continuously depending on the grid operator's requirements and the capabilities of the installed equipment. Active power can be controlled as a function of frequency and the reactive power as a function of voltage or active power following the grid operator's characteristic curves. Ramp rate limitations are enforced during both active power control functions as well normal plant operations. Reactive power compensation of passive components and reactive power injection into the grid is supported using fast or slow reactive compensation components.

Closed-loop PID control algorithms integrated with feed-forward techniques ensure high accuracy and fast response time:

- High accuracy in PPC-based active and reactive power control is ensured through intelligent closed loop PID control algorithms taking into account grid code requirements and limitations.
- Fast PID response time is ensured through proper control loop tuning and accelerated through knowledge of the plant system performance and losses, integrating a feed forward mechanism.

4.Network Set-up

4.1 IP Addressing Table

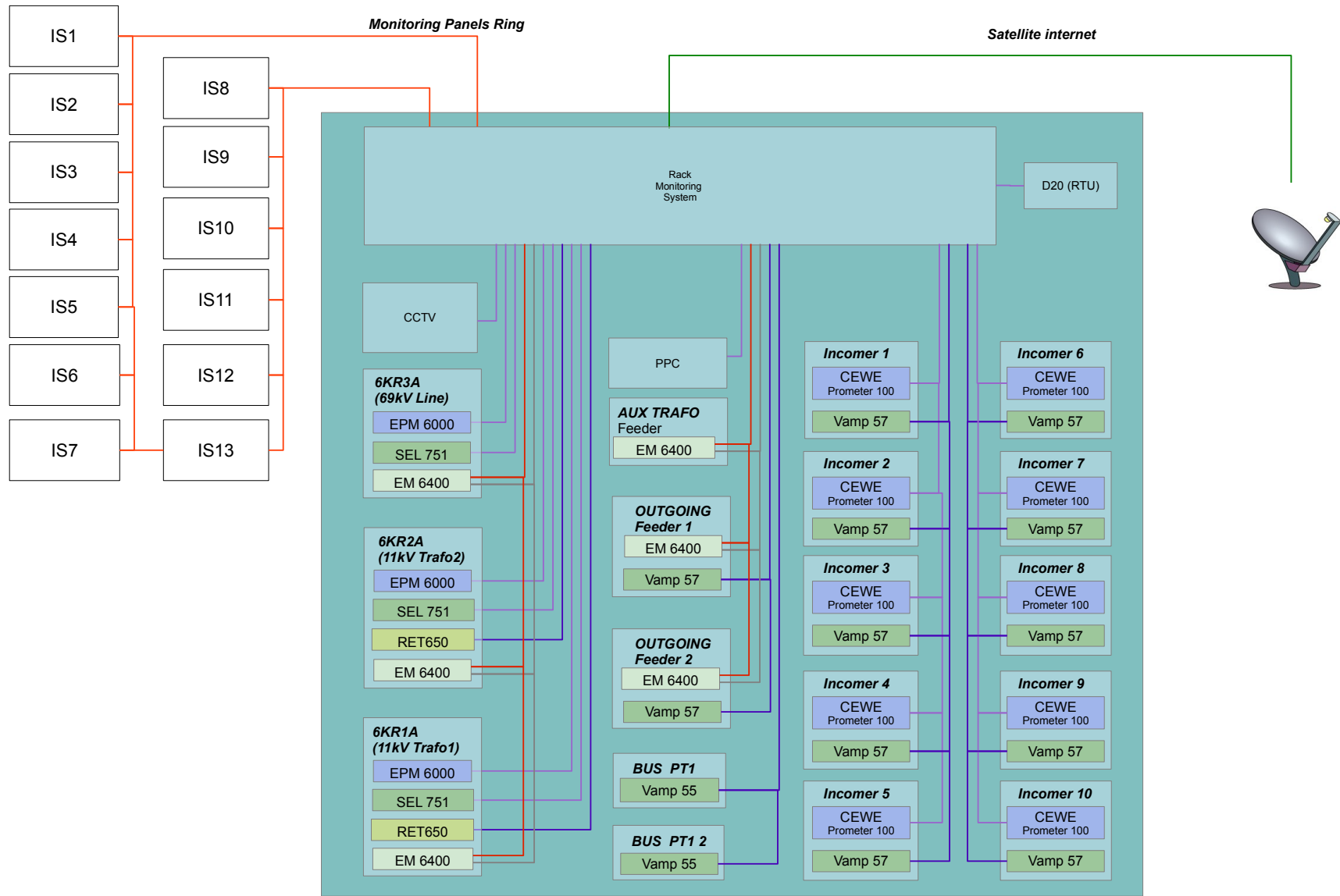
#	Device	IP	Subnet	Gateway	Comments
Main S/S					
	Main Router	192.168.1.1	255.255.255.0		MAIN ROUTER
1	LPS Server	192.168.1.10	255.255.255.0	192.168.1.1	
2	Back UP Server	192.168.20.9	255.255.255.0	192.168.1.1	
3	Desktop SCADA	192.168.1.15	255.255.255.0	192.168.1.1	
4	Mikrotik RB750	192.168.1.99 192.168.20.99	255.255.255.0	192.168.1.1	IPSEC GATEWAY
5	RSC10 Controller 0	192.168.1.100 192.168.20.100	255.255.255.0	192.168.20.99	INDUSTRIAL PC
6	RSC10 Controller 15	192.168.1.115 192.168.20.115	255.255.255.0	192.168.20.99	INDUSTRIAL PC
7	SISTM1013 Optical	192.168.20.20	255.255.255.0	192.168.20.99	Managed Switch
8	Energy Meter I/C 1	192.168.1.160	255.255.255.0	192.168.1.1	Meter
9	Energy Meter I/C 2	192.168.1.161	255.255.255.0	192.168.1.1	Meter
10	Energy Meter I/C 3	192.168.1.162	255.255.255.0	192.168.1.1	Meter
11	Energy Meter I/C 4	192.168.1.163	255.255.255.0	192.168.1.1	Meter
12	Energy Meter I/C 5	192.168.1.164	255.255.255.0	192.168.1.1	Meter
13	Energy Meter I/C 6	192.168.1.165	255.255.255.0	192.168.1.1	Meter
14	Energy Meter I/C 7	192.168.1.166	255.255.255.0	192.168.1.1	Meter
15	Energy Meter I/C 8	192.168.1.167	255.255.255.0	192.168.1.1	Meter
16	Energy Meter I/C 9	192.168.1.168	255.255.255.0	192.168.1.1	Meter
17	Energy Meter I/C 10	192.168.1.169	255.255.255.0	192.168.1.1	Meter
18	Energy Meter Trafo 1	192.168.1.170	255.255.255.0	192.168.1.1	Meter
19	Energy Meter Trafo 2	192.168.1.171	255.255.255.0	192.168.1.1	Meter
20	Energy Meter 69kV Line	192.168.1.172	255.255.255.0	192.168.1.1	Meter
21	Protection Relay I/C 1	192.168.1.173	255.255.255.0	192.168.1.1	Relay
22	Protection Relay I/C 2	192.168.1.174	255.255.255.0	192.168.1.1	Relay
23	Protection Relay I/C 3	192.168.1.175	255.255.255.0	192.168.1.1	Relay
24	Protection Relay I/C 4	192.168.1.176	255.255.255.0	192.168.1.1	Relay
25	Protection Relay I/C 5	192.168.1.177	255.255.255.0	192.168.1.1	Relay
26	Protection Relay I/C 6	192.168.1.178	255.255.255.0	192.168.1.1	Relay
27	Protection Relay I/C 7	192.168.1.179	255.255.255.0	192.168.1.1	Relay
28	Protection Relay I/C 8	192.168.1.180	255.255.255.0	192.168.1.1	Relay
29	Protection Relay I/C 9	192.168.1.181	255.255.255.0	192.168.1.1	Relay
30	Protection Relay I/C 10	192.168.1.182	255.255.255.0	192.168.1.1	Relay
31	Protection Relay Outgoing 1	192.168.1.183	255.255.255.0	192.168.1.1	Relay
32	Protection Relay Outgoing 2	192.168.1.184	255.255.255.0	192.168.1.1	Relay
33	Protection Relay Bus PT1	192.168.1.185	255.255.255.0	192.168.1.1	Relay
34	Protection Relay Bus PT2	192.168.1.186	255.255.255.0	192.168.1.1	Relay
35	Relay Trafo 1	192.168.1.189	255.255.255.0	192.168.1.1	Relay
36	Relay Trafo 2	192.168.1.190	255.255.255.0	192.168.1.1	Relay
37	Relay Power Trafo 1	192.168.1.191	255.255.255.0	192.168.1.1	Relay
38	Relay Power Trafo 2	192.168.1.192	255.255.255.0	192.168.1.1	Relay
39	Relay 69kV Line	192.168.1.193	255.255.255.0	192.168.1.1	Relay
40	GPS Server	192.168.1.198	255.255.255.0	192.168.1.1	GPS Server
41	D 20 -1st IP	192.168.1.199	255.255.255.0	192.168.1.1	RTU
42	D 20 -2nd IP	192.168.1.42	255.255.255.0	192.168.1.1	RTU
43	D 20 -3th IP	192.168.1.43	255.255.255.0	192.168.1.1	RTU
44	D 20 -4th IP	192.168.1.44	255.255.255.0	192.168.1.1	RTU
S/S1					

#	Device	IP	Subnet	Gateway	Comments
45	Industrial PC 1	192.168.1.101 192.168.20.101	255.255.255.0	192.168.20.99	INDUSTRIAL PC
46	SISTM1013 Optical	192.168.20.21	255.255.255.0	192.168.20.99	Managed Switch
47	Expert 9132	192.168.20.50	255.255.255.0	192.168.20.99	Converter Ether2 Serial
48	Inverter 1	192.168.1.131	255.255.255.0	192.168.1.1	
49	Inverter 1	192.168.1.222	255.255.255.0	192.168.1.1	
50	Inverter 2	192.168.1.132	255.255.255.0	192.168.1.1	
51	Inverter 2	192.168.1.223	255.255.255.0	192.168.1.1	
S/S2					
52	Industrial PC 2	192.168.1.102 192.168.20.102	255.255.255.0	192.168.20.99	INDUSTRIAL PC
53	SISTM1013 Optical	192.168.20.22	255.255.255.0	192.168.20.99	Managed Switch
54	Expert 9132	192.168.20.51	255.255.255.0	192.168.20.99	Converter Ether2 Serial
55	Inverter 3	192.168.1.133	255.255.255.0	192.168.1.1	
56	Inverter 3	192.168.1.224	255.255.255.0	192.168.1.1	
57	Inverter 4	192.168.1.134	255.255.255.0	192.168.1.1	
58	Inverter 4	192.168.1.225	255.255.255.0	192.168.1.1	
S/S3					
59	Industrial PC 3	192.168.1.103 192.168.20.103	255.255.255.0	192.168.20.99	INDUSTRIAL PC
60	SISTM1013 Optical	192.168.20.23	255.255.255.0	192.168.20.99	Managed Switch
61	Expert 9132	192.168.20.52	255.255.255.0	192.168.20.99	Converter Ether2 Serial
62	Inverter 5	192.168.1.135	255.255.255.0	192.168.1.1	
63	Inverter 5	192.168.1.226	255.255.255.0	192.168.1.1	
64	Inverter 6	192.168.1.136	255.255.255.0	192.168.1.1	
65	Inverter 6	192.168.1.227	255.255.255.0	192.168.1.1	
65S/S4					
66	Industrial PC 4	192.168.1.104 192.168.20.104	255.255.255.0	192.168.20.99	INDUSTRIAL PC
67	SISTM1013 Optical	192.168.20.24	255.255.255.0	192.168.20.99	Managed Switch
68	Expert 9132	192.168.20.53	255.255.255.0	192.168.20.99	Converter Ether2 Serial
69	Inverter 7	192.168.1.137	255.255.255.0	192.168.1.1	
70	Inverter 7	192.168.1.228	255.255.255.0	192.168.1.1	
71	Inverter 8	192.168.1.138	255.255.255.0	192.168.1.1	
72	Inverter 8	192.168.1.229	255.255.255.0	192.168.1.1	
S/S5					
73	Industrial PC 5	192.168.1.105 192.168.20.105	255.255.255.0	192.168.20.99	INDUSTRIAL PC
74	SISTM1013 Optical	192.168.20.25	255.255.255.0	192.168.20.99	Managed Switch
75	Expert 9132	192.168.20.54	255.255.255.0	192.168.20.99	Converter Ether2 Serial
76	Inverter 9	192.168.1.139	255.255.255.0	192.168.1.1	
77	Inverter 9	192.168.1.230	255.255.255.0	192.168.1.1	
78	Inverter 10	192.168.1.140	255.255.255.0	192.168.1.1	
79	Inverter 10	192.168.1.231	255.255.255.0	192.168.1.1	
S/S6					
80	Industrial PC 6	192.168.1.106 192.168.20.106	255.255.255.0	192.168.20.99	INDUSTRIAL PC
81	SISTM1013 Optical	192.168.20.26	255.255.255.0	192.168.20.99	Managed Switch
82	Expert 9132	192.168.20.55	255.255.255.0	192.168.20.99	Converter Ether2 Serial
83	Inverter 11	192.168.1.141	255.255.255.0	192.168.1.1	
84	Inverter 11	192.168.1.232	255.255.255.0	192.168.1.1	
85	Inverter 12	192.168.1.142	255.255.255.0	192.168.1.1	
86	Inverter 12	192.168.1.233	255.255.255.0	192.168.1.1	
87	Weather Station 1	192.168.1.195	255.255.255.0	192.168.1.1	Weather Data Logger
S/S3					
88	Industrial PC 7	192.168.1.107 192.168.20.107	255.255.255.0	192.168.20.99	INDUSTRIAL PC

#	Device	IP	Subnet	Gateway	Comments
89	SISTM1013 Optical	192.168.20.27	255.255.255.0	192.168.20.99	Managed Switch
90	Expert 9132	192.168.20.56	255.255.255.0	192.168.20.99	Converter Ether2 Serial
91	Inverter 13	192.168.1.143	255.255.255.0	192.168.1.1	
92	Inverter 13	192.168.1.234	255.255.255.0	192.168.1.1	
93	Inverter 14	192.168.1.144	255.255.255.0	192.168.1.1	
94	Inverter 14	192.168.1.235	255.255.255.0	192.168.1.1	
S/S8					
95	Industrial PC 8	192.168.1.108 192.168.20.108	255.255.255.0	192.168.20.99	INDUSTRIAL PC
96	SISTM1013 Optical	192.168.20.28	255.255.255.0	192.168.20.99	Managed Switch
97	Expert 9132	192.168.20.57	255.255.255.0	192.168.20.99	Converter Ether2 Serial
98	Inverter 15	192.168.1.145	255.255.255.0	192.168.1.1	
99	Inverter 15	192.168.1.236	255.255.255.0	192.168.1.1	
100	Inverter 16	192.168.1.146	255.255.255.0	192.168.1.1	
101	Inverter 16	192.168.1.237	255.255.255.0	192.168.1.1	
S/S9					
102	Industrial PC 9	192.168.1.109 192.168.20.109	255.255.255.0	192.168.20.99	INDUSTRIAL PC
103	SISTM1013 Optical	192.168.20.29	255.255.255.0	192.168.20.99	Managed Switch
104	Expert 9132	192.168.20.58	255.255.255.0	192.168.20.99	Converter Ether2 Serial
105	Inverter 17	192.168.1.147	255.255.255.0	192.168.1.1	
106	Inverter 17	192.168.1.238	255.255.255.0	192.168.1.1	
107	Inverter 18	192.168.1.148	255.255.255.0	192.168.1.1	
108	Inverter 18	192.168.1.239	255.255.255.0	192.168.1.1	
S/S10					
109	Industrial PC 10	192.168.1.110 192.168.20.110	255.255.255.0	192.168.20.99	INDUSTRIAL PC
110	SISTM1013 Optical	192.168.20.30	255.255.255.0	192.168.20.99	Managed Switch
111	Expert 9132	192.168.20.59	255.255.255.0	192.168.20.99	Converter Ether2 Serial
112	Inverter 19	192.168.1.149	255.255.255.0	192.168.1.1	
113	Inverter 19	192.168.1.240	255.255.255.0	192.168.1.1	
114	Inverter 20	192.168.1.150	255.255.255.0	192.168.1.1	
115	Inverter 20	192.168.1.241	255.255.255.0	192.168.1.1	
S/S11					
116	Industrial PC 11	192.168.1.111 192.168.20.111	255.255.255.0	192.168.20.99	INDUSTRIAL PC
117	SISTM1013 Optical	192.168.20.31	255.255.255.0	192.168.20.99	Managed Switch
118	Expert 9132	192.168.20.60	255.255.255.0	192.168.20.99	Converter Ether2 Serial
119	Inverter 21	192.168.1.151	255.255.255.0	192.168.1.1	
120	Inverter 21	192.168.1.242	255.255.255.0	192.168.1.1	
121	Inverter 22	192.168.1.152	255.255.255.0	192.168.1.1	
122	Inverter 22	192.168.1.243	255.255.255.0	192.168.1.1	
123	Inverter 23	192.168.1.153	255.255.255.0	192.168.1.1	
124	Inverter 23	192.168.1.244	255.255.255.0	192.168.1.1	
S/S12					
125	Industrial PC 12	192.168.1.112 192.168.20.112	255.255.255.0	192.168.20.99	INDUSTRIAL PC
126	SISTM1013 Optical	192.168.20.32	255.255.255.0	192.168.20.99	Managed Switch
127	Expert 9132	192.168.20.61	255.255.255.0	192.168.20.99	Converter Ether2 Serial
128	Inverter 24	192.168.1.154	255.255.255.0	192.168.1.1	
129	Inverter 24	192.168.1.245	255.255.255.0	192.168.1.1	
130	Inverter 25	192.168.1.155	255.255.255.0	192.168.1.1	
131	Inverter 25	192.168.1.246	255.255.255.0	192.168.1.1	
132	Weather Station 2	192.168.1.196	255.255.255.0	192.168.1.1	Weather Data Logger
S/S13					
133	Industrial PC 13	192.168.1.113 192.168.20.113	255.255.255.0	192.168.20.99	INDUSTRIAL PC
134	SISTM1013 Optical	192.168.20.33	255.255.255.0	192.168.20.99	Managed Switch

#	Device	IP	Subnet	Gateway	Comments
135	Expert 9132	192.168.20.62	255.255.255.0	192.168.20.99	Converter Ether2 Serial
136	Inverter 26	192.168.1.156	255.255.255.0	192.168.1.1	
137	Inverter 26	192.168.1.247	255.255.255.0	192.168.1.1	
138	Inverter 27	192.168.1.157	255.255.255.0	192.168.1.1	
139	Inverter 27	192.168.1.248	255.255.255.0	192.168.1.1	
S/S14 - PPC					
140	Industrial PC 14	192.168.1.114 192.168.20.114	255.255.255.0	192.168.20.99	INDUSTRIAL PC
CCTV					
141-152	Cameras - CCTV IP's	192.168.1.201-221	255.255.255.0	192.168.1.1	

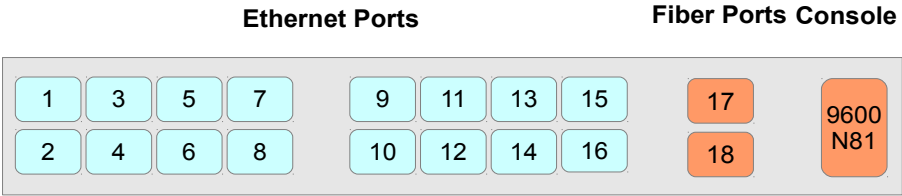
4.2 Communication Layout



 inaccess S.A Sorou 12, Marousi Athens 15125 Tel: +30 2106802300 Fax: +30 2106199258 www.inaccess.com	To: Sterling & Wilson	Drawing Title: Philippines Digos 28.59MWp	UTP Cable Cat5e Ground Cable RS232 Cable Modbus TCP Cable IEC 61850 TCP Cable 0.25mm Cable RS485 D+/- 0.25mm DigitalContact 0.25mm +/-24VDC Optic Fiber Patch Cord  LEGEND
	Project Name & Address: Philippines Digos 28.59MWp	Date & Version: 14-07-2015 V 1.0.0	

4.3 Switch and Router Layout

Transition Switch



1. MikroTik RB 750

2. Bridge SW to ESW

3. Controller RSC 10-110 1

4. Controller RSC 10-110 2

5. Main LPS Server

6. Main LPS Server ILO

7. Back Up Server LPS

8. Back Up Server LPS ILO
9.

10.

11.

12.

13.

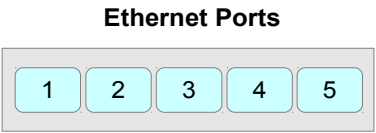
14.

15.

16.
17. Monitoring Ring from IS1

18. Monitoring Ring from IS13

Mikrotik RB 750



1. Do not use

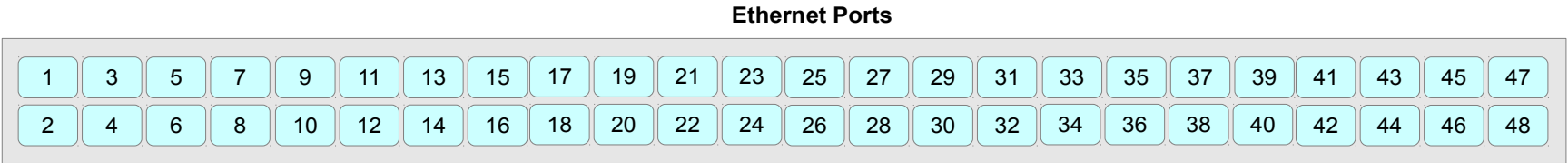
2. Internet

3. Controller

4. DVR

5. Spare

Ethernet Switch



1. PPC

2. D20 (RTU)

3. NTP Server

4.

5.

6.

7. CCTV Alarm

8. CCTV PC

9.

10.

11.

12.
- 13 Incomers Meter CEWE

14. Incomers Relay Vamp57

15. Outgoers Relay Vamp57

16. Bus PT Relay Vamp55

17.

18.

19.

20.

21.

22.

23.

24.
25. AKR1A Meter EMP 6000

26. AKR1A Relay SeL 751

27. AKR1A Relay RET 650

28. AKR2A Meter EMP 6000

29. AKR2A Relay SeL 751

30. AKR2A Relay RET 650

31. AKR3A Meter EMP 6000

32. AKR3A Relay SeL751

33.

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47.

48. Bridge SW1 & SW2